Paper / Subject Code: 32405 / Elective - I Advance Data Structures & Analysis of Algorithms (DLOC)

TE / Sem V / CBCGS / IT / ND-18 / 12-12-2018

(3 Hours) **[Total Marks: 80]**

N.B.: (1) Question No.1 is compulsory.

(2) Attempt **any three** out of remaining questions.

(3) Assume Suitable data if necessary.

(4) **Figures** to the **right** indicate full **marks**.

Q1.	(a)	Explain with example how divide and conquer strategy is used in Binary Search?	5
	(b)	Explain flow shop scheduling technique.	5
	(c)	Write a note on AVL Tree.	5
	(d)	Write an algorithm for finding minimum and maximum number from given set.	5

Q2. (a) What is longest common subsequence problem? Find LCS for following string. 10

X=ACBAED

Y=ABCABE

(b) Which are the different methods of solving recurrences? Explain with examples. 10

Q3. (a) Compare Greedy and Dynamic Programming approach for an algorithm design. Explain how both can be used to solve knapsack problem. 10

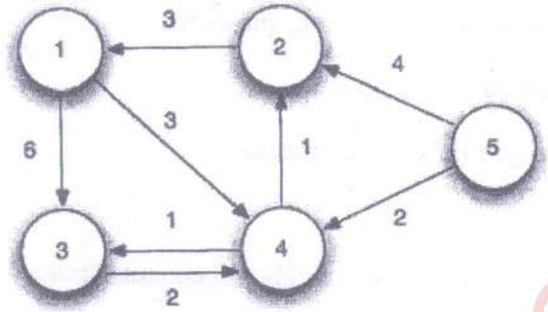
(b) Explain Huffman algorithm. Construct Huffman tree for **MAHARASHTRA** with its optimal code. 10

Q4. (a) Explain Job sequencing with deadlines. 10
Let n=4,(p1,p2,p3,p4)=(100,10,15,27) and (d1,d2,d3,d4)=(2,1,2,1).Find feasible solution.

(b) Sort the following numbers using quick sort. Also derive time complexity of quick sort. 10

27 10 36 18 25 45

Q5. (a) Apply all pair shortest path on the following graph 10



(b) Given a chain of four matrices A_1, A_2, A_3 and A_4 with $P_0=5$, $P_1=4$, $P_2=6$, $P_3=2$ and $P_4=7$. Find m[1,4] using matrix chain multiplication 10

Q6. Write Note on (**Any two**) 20

i. Rabin Karp Algorithm.
ii. Topological Sort.
iii. Knuth-Morrie-Pratt algorithm.
iv. Red-Black Tree.

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(3 Hours) [Marks: 80]

N.B.: 1) Question No. 1 is compulsory.
2) Answer any three out of remaining questions.
3) Assume suitable data if necessary.
4) Figures to the right indicate full marks.

Q1. (a) Compute the worst case complexity of the following program segment: (5)

```
void fun(int n, int arr[]){
 int i = 0, j = 0;
for(; i < n; ++i)
while(j < n && arr[i] < arr[j])
j++;
}
```

(b) Differentiate between greedy method and dynamic programming? (5)

(c) . What is the optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers? (5)

a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21

(d) Find Longest Common Subsequence for the following: (5)

String x=ACBAED
String y=ABCABE

Q2. (a)Consider the instance of knapsack problem where n=6, M=15, profits are (P1,P2,P3,P4,P5,P6) =(1,2,4,4,7,2) and weights are (W1,W2,W3,W4,W5,W6)=(10,5,4,2,7,3). Find maximum profit using fractional Knapsack. (10)

(b) Explain divide and conquer approach. Write a recursive algorithm to determine the max and min from given elements. (10)

Q3. (a) Define AVL tree. Construct AVL tree for the following data: (10)

21,26,30,9,4,14,28,18,15,10,2,3,7

(b) A traveler needs to visit all the cities from a list (figure 1), where distances between all the cities are known and each city should be visited just once. What is the shortest possible route that he visits each city exactly once and returns to the origin city? (10)

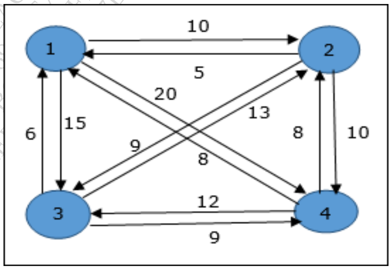

Figure 1.

Q4. (a) Construct a minimum spanning tree shown in figure 2 using Kruskal's and Prim's Algorithm and find out the cost with all intermediate steps. (10)

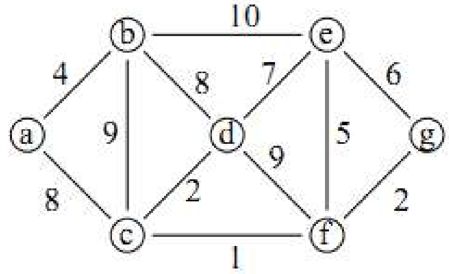

Figure 2

(b) What is optimal binary search tree? Explain with the help of example. (10)

Q5. (a) Give asymptotic upper bound for T(n) for the following recurrences and verify your answer using Masters theorem:

$$T(n) = T(n-1) + n \qquad (10)$$

(b) Given a set of 9 jobs (J1,J2,J3,J4,J5,J6,J7,J8,J9) where each job has a deadline (5,4,3,3,4,5,2,3,7) and profit (85,25,16,40,55,19,92,80,15) associated to it .Each job takes 1 unit of time to complete and only one job can be scheduled at a time. We earn the profit if and only if the job is completed by its deadline. The task is to find the maximum profit and the number of jobs done. (10)

Q6. Explain any Two: (20)

a) Rabin Karp Algorithm
b) Genetic Algorithm
c) Minimum Cost Spanning Tree
d) Red Black Trees

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1T01233 - S.E.(Information Technology Engineering)(SEM-III)(Choice Base Credit Grading System) (R- 19) (C Scheme) / 51422 - Data Structure & Analysis
QP CODE: 10030183 DATE: 26/05/2023

(3 Hours) **[Marks: 80]**

N.B.: 1) Question No. 1 is compulsory.
2) Answer any three out of remaining questions.
3) Assume suitable data if necessary.
4) Figures to the right indicate full marks.

Q1. (a) Explain Linked lists in detail (5)
(b) List down the applications of stack. (5)
(c) Explain winding and unwinding phase of recursion. (5)
(d) Briefly explain memory fragmentation. (5)

Q2. (a) Design an algorithm to implement circular queue using an array. (10)
Q2. (b)Explain quick sort with example by giving its algorithm and comment on its complexity. (10)

Q3. (a) Write an algorithm to covert infix expression to postfix expression. (10)
Q3. (b) Explain various collision resolution techniques in hashing. (10)

Q4. (a) Define Minimum Spanning Tree. Construct a minimum spanning tree shown in figure 1 using Kruskal's and Prim's Algorithm and find out the cost with all intermediate steps. (10)

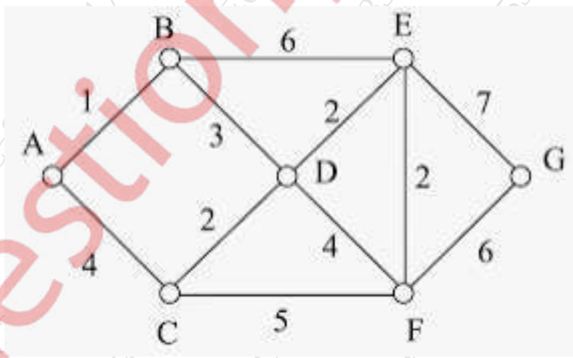

Figure 1

Q4. (b) Define AVL Tree. Step by step construct a AVL tree for the following data
23, 12, 25, 01, 45,63, 27, 29 ,90,78,5,6,10 (10)

Q5. (a) Write down the algorithm for addition of two polynomials. (10)
Q5.(b) Define Binary Search Tree. Give the algorithms for various tree traversals. (10)

Q6. Solve **any Four:** (20)

a) Threaded Binary Tree
b) Depth First Search
c) Game Tree
d) Selection Sort
e) B+-tree

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(3 Hours) **[Total Marks: 80]**

N.B.: (1) Question No.1 is compulsory.

(2) Attempt **any three** out of remaining questions.

(3) Assume Suitable data if necessary.

(4) **Figures** to the **right** indicate full **marks**.

Q1 a. Differentiate between Greedy method and Dynamic Programming. 5

b. Write an algorithm for finding minimum and maximum number from a given set 5

c. Explain coin changing problem 5

d. Explain Flow Shop Scheduling Technique 5

Q2a. Define AVL tree. Construct an AVL tree for the following data. 10

63, 9, 19, 27, 18, 108, 99, 81

b. Write an algorithm for implementing Quick sort. Also, comment on its complexity. 10

Q3a. What is longest common subsequence problem? Find LCS for the following string: 10

String X: ABCDGH

String Y: AEDFHR

b. Explain Rabin Karp Algorithm in detail. 10

Q4a. Which are the different methods of solving recurrences? Explain with suitable examples. 10

b. Explain Travelling Salesman Problem with an example. 10

Q5a. Explain Huffman Algorithm. Construct a Huffman Tree and find Huffman code for the message: KARNATAKA. 10

b. Explain Knapsack Problem with an example. 10

Q6 Write Short notes on (**any four**) 20

a. Genetic Algorithm
b. Red and Black Tree
c. Merge Sort
d. Knuth Morris Pratt Algorithm
e. Optimal Binary Search Tree (OBST)

(3 hours) **(Marks : 80)**

N.B.: (1) Question No. 1 is compulsory.
(2) Attempt any three out of the remaining five questions.
(3) Assumptions made should be clearly stated.

Q1 Answer the following Any Four.

a) What is Complexity? Explain in detail asymptotic notations. **5**
b) Explain approximation algorithms with an example. **5**
c) Compare Greedy approach and Dynamic Programming approach for an algorithm design. **5**
d) Describe naive string matching method. Write the algorithm for the same. **5**
e) Build a max heap for the following. 45, 65, 34, 25, 78, 56, 15. **5**

Q2

a) Define B-tree. Explain insertion and deletion operations on a B tree, with an example of each. **10**
b) Differentiate between Prims and Kruskals algorithms **10**

Q3

a) Find the longest common subsequence for the following two strings, using dynamic programming. X=abcabcba, Y= babcbcab **10**
b) Which are the different methods of solving recurrences. Explain with examples **10**

Q4

a) Consider the instance of knapsack problem where n=6, M=15, profits are (P1,P2,P3,P4,P5,P6) = (1,2,4,4,7,2) and weights are (W1,W2,W3,W4,W5,W6)= (10,5,4,2,7,3). Find maximum profit using Fractional knapsack. **10**
b) Explain matrix chain multiplication in detail. **10**

Q5

a) Sort the following numbers using Quicksort algorithm. 20, 30, 14, 56, 9, 72, 45, 5. **10**
b) Describe, with the help of an example, KMP algorithm. Also, comment on complexity. **10**

Q6.

a) Explain genetic algorithms in detail. **10**
b) Write a note on optimal binary search tree. **10**

1T01233 - S.E.(Information Technology Engineering)(SEM-III)(Choice Base Credit Grading System) (R-2020-21) (C Scheme) / 51422 - Data Structure & Analysis

QP CODE:10012367 DATE: 23/11/2022

(3 Hours) [Marks: 80]

N.B.: 1) Question No. 1 is compulsory.
2) Answer any three out of remaining questions.
3) Assume suitable data if necessary.
4) Figures to the right indicate full marks.

Q1. (a) Define Graph. List down the applications of graph. (5)
(b) Explain winding and unwinding phase of recursion. (5)
(c) Explain the concept of Buddy system. (5)
(d) Explain different types of link list. (5)

Q2. (a) Design an algorithm to perform the following operations on circular link list: (10)
i) Insert node at the end
ii) Delete the first node
iii)Count number of nodes

Q2. (b)Explain selection sort with example by giving its algorithm and comment on its complexity. (10)

Q3. (a) Construct a B-tree of order 5 for the following set of data: (10)
1,12,8,2,25,6,14,28,17,7,52,16,48,68,3,26,29,53,55,45,67

Q3. (b) Write an algorithm for Breadth First Search (BFS) and traverse the graph shown in figure 1 using BFS. (10)

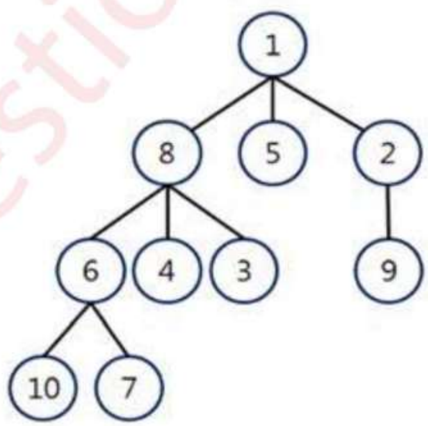

Figure 1

Q.4. (a) Write a program to implement Stack ADT using Linked list. (10)

Q.4. (b) Explain in brief: - (10)

i. Directed Graph
ii. Weighted Graph
iii. Minimum Spanning Tree
iv. Adjacency Matrix Representation
v. Adjacency list Representation.

Q.5 (a) Write a program to implement queue using array. (10)

Q.5 (b) Sort the following data in ascending order using Heap Sort. 20,14,50,3,5,7,11,8,12,15. Show all the steps. Write an algorithm for Heap Sort.

Q.6.Write Short note on any four: - (20)

i. AVL Tree
ii. Circular Queue
iii. Binary Search
iv. Hashing Techniques
v. Dijkstra's Algorithm

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University of Mumbai

Examinations Summer 2022

Time: 2 hour 30 minutes Max. Marks: 80

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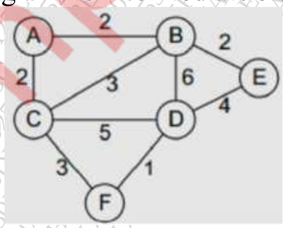
Q1.	Choose the correct option for following questions. All the Questions are compulsory and carry equal marks
1.	Process of inserting an element in stack is called
Option A:	Create
Option B:	Push
Option C:	Evaluation
Option D:	Pop
2.	Consider the usual algorithm for determining whether a sequence of parentheses is balanced. The maximum number of parentheses that appear on the stack AT ANY ONE TIME when the algorithm analyzes: (()(())(()))?
Option A:	1
Option B:	2
Option C:	3
Option D:	4 or more
3.	Which of the following statements is true?
Option A:	Recursion is always better than iteration
Option B:	Recursion uses more memory compared to iteration
Option C:	Recursion uses less memory compared to iteration
Option D:	Iteration is always better and simpler than recursion
4.	The number of elements in the adjacency matrix of a graph having 7 vertices is
Option A:	7
Option B:	14
Option C:	36
Option D:	49
5.	In a max-heap, element with the greatest key is always in the which node?
Option A:	Leaf node
Option B:	First node of left sub tree
Option C:	root node
Option D:	First node of right sub tree
6.	________________ can be found used to find a minimum spanning tree.
Option A:	Prim's Algorithm
Option B:	Breadth First
Option C:	Dijkstra's Algorithm
Option D:	Flloyd Warshal Algorithm
7.	Which data structure is required to evaluate a postfix expression
Option A:	Stack
Option B:	Queue
Option C:	Array
Option D:	Linked−list
8.	A binary tree in which if all its levels except possibly the last, have the maximum number of nodes and all the nodes at the last level appear as far left as possible, is called
Option A:	Threaded tree

Option B:	Full binary tree
Option C:	Binary Search Tree
Option D:	Complete binary tree
9.	The number of edges from the root to the node is called __________ of the tree.
Option A:	Height
Option B:	Depth
Option C:	Length
Option D:	Width
10.	Which of the following is not a collision resolution technique?
Option A:	Rehashing
Option B:	Clustering
Option C:	Linear Probing
Option D:	Quadratic Probing

Q2. (20 Marks)	**Solve any Two Questions out of Three** **10 marks each**
A	Explain Doubly ended queue.Explain the variants of Doubly ended queue.
B	Explain BFS algorithm using an example of your own.
C	Write an algorithm to implement circular linked list.

Q3. (20 Marks)	**Solve any Two Questions out of Three** **10 marks each**
A	Find the Minimum spanning tree for the graph shown if figure1 using Prim's and Kruskal's algorithm by showing all the intermediate steps.  Figure 1. Graph
B	Explain different collision resolution techniques. Insert the following sequence of keys in the hash table with a size of 10 using linear probing**{18,89,21,58,68,11}**
C	Explain Heap sort with the help of an example.

Q4. (20 Marks)	**Solve any Two Questions out of Three** **10 marks each**
A	What are different tree traversal methods? Create a binary search tree for 16,70,10,30,75,5,12,9 and traverse the tree in in-order, pre-order and post-order.
B	Create a B-tree of order 4 with the following keys: 60,70,75,51,52,65,68,77,78,79
C	Write an algorithm to convert infix to postfix expression.

1T01233 - S.E.(Information Technology Engineering)(SEM-III)(Choice Base Credit Grading System) (R-19) (C Scheme) / 51422 - Data Structure & Analysis

QP CODE: 10038221 DATE: 23/11/2023

(3 Hours) **[Marks: 80]**

N.B.: 1) Question No. 1 is compulsory.
2) Answer any three out of remaining questions.
3) Assume suitable data if necessary.
4) Figures to the right indicate full marks.

Q1. (a) Explain data structures and Abstract Data Type. (5)
(b) What is expression tree? Give examples. (5)
(c) What is a Linked List? State the different types of Linked List. (5)
(d) What are the different ways to represent Graph. (5)

Q2. (a) Write an algorithm to implement queue using an array. (10)

Q2. (b)Explain insertion sort with example by giving its algorithm and comment on its complexity. (10)

Q3. (a) Write an algorithm to implement stack using array. (10)

Q3. (b) What is Doubly Linked List? Write an algorithm to implement following operations on Doubly Linked List.
a) Insertion (all cases)
b) Traversal (Forward and Backward) (10)

Q4. (a) Define Minimum Spanning Tree. Construct a minimum spanning tree shown in figure 1 using Kruskal's and Prim's Algorithm and find out the cost with all intermediate steps. (10)

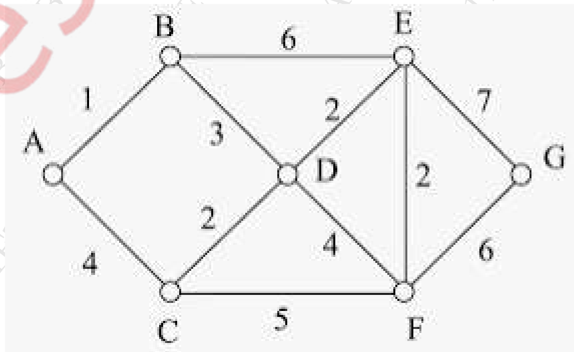


Figure 1

Q4. (b) Define AVL tree. Construct an AVL tree from the following data and mention the rotations in each step. (10)

40,30,20,25,21,50,60,70,65,22,18,15

Q5. (a) What is collision? List down the methods to resolve the collision. Consider a hash table of size 11. Using linear probing, insert keys 54, 26, 93, 17, 77, 60 and 31 into the table. (10)

Q5. (b) Write the algorithm for deletion of a node (all cases) in a Binary Search Tree. (10)

Q6. Write Short note on any **four**: (20)

a) Breadth First Search
b) Expression Tree
c) Selection Sort
d) Double Ended Queue (De-Queue)
e) Binary Search

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Paper / Subject Code: 40522 / Analysis of Algorithm

(3 Hours) **Total Marks: 80**

N.B: (1) Question No. 1 is compulsory.
(2) Attempt any three from the remaining questions.
(3) Figures to the right indicate full marks.

1. Attempt any four
   (a) Explain Best Case, Average Case and Worst Case. **(05)**
   (b) Explain Multistage graphs. **(05)**
   (c) Explain Binary search algorithm. **(05)**
   (d) Define NP Class, NP hard, NP complete. **(05)**
   (e) What is greedy algorithm? **(05)**

2.
   (a) Write and explain sum of subset algorithm for n=5, W= {2,7,8,9,15}, M=17. **(10)**

   (b) Obtain the solution to the following knapsack problem using Greedy method: n=7, m=15 (p1,p2.....p7) = (10,5,15,7,6,18,3), (w1,w2,.....,w7) = (2,3, 5,7,1,4,1). **(10)**

3.
   (a) What is the Longest Common Subsequence problem? Find the LCS for following strings **(10)**

   String 1- ACBAED
   String 2- ABCABE

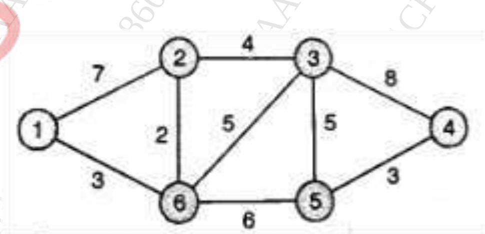
   (b) Explain quick sort with algorithm and example. **(10)**

4.
   (a)What is Knuth Morris Pratt Method of Pattern Matching? Give Examples. **(10)**

   (b) Solve the following Recurrence using Substitution Method. **(10)**

$$T(n) = \begin{cases} 1, & \text{if } n=1 \\ 2\,T(n/2) + Cn, & \text{if } n>1 \end{cases}$$

5.
   (a) Find the Dijkstra's shortest path from vertex 1 to vertex 4 for the following graph. **(10)**



   (b) Apply Merge sort algorithm to sort the following numbers. Show each step clearly. 10, 5, 7, 6, 1, 4, 8, 3, 2, 9. **(10)**

6. Write notes on **(any two):** **(20)**

   (a) Find Minimum and Maximum elements of an array X[0 : 9] = (45, 83, 75, 17, 43, 37, 80, 53, 61, 22) using divide and conquer strategy.
   (b) Naïve string matching algorithm with example.
   (c) N-queen problem algorithm with example.

15407

SE(COMP) / SEM-IV / R-19 / 20.05.22(E) / AOA

QP CODE: 93328

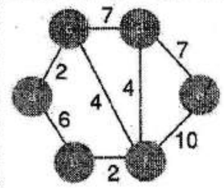
University of Mumbai

Examinations Summer 2022

Time: 2 hour 30 minutes | Max. Marks: 80

Q1. (20 Marks)	**Choose the correct option for following questions. All the Questions are compulsory and carry equal marks**
1.	_________ is the class of decision problems that can be solved by non-deterministic polynomial algorithms.
Option A:	NP
Option B:	P
Option C:	Hard
Option D:	Complete
2.	Following data structure is used to implement LIFO Branch and Bound Strategy
Option A:	Priority Queue
Option B:	array
Option C:	stack
Option D:	Linked list
3.	For the given elements 6 4 11 17 2 24 14 using quick sort, what is the sequence after first phase, assuming the pivot as the first element?
Option A:	2 4 6 17 11 24 14
Option B:	2 4 6 11 17 14 24
Option C:	4 2 6 17 11 24 14
Option D:	2 4 6 11 17 24 14
4.	Which of the following is correct for branch and bound technique? i. It is BFS generation of problem states ii. It is DFS generation of problem states iii. It is D-search.
Option A:	Only i
Option B:	Only ii
Option C:	Only ii and iii
Option D:	Only i, and iii
5.	Consider the given graph.

64A6AB6403A4A6CB48000E1337C99516

	 What is the weight of the minimum spanning tree using the Kruskal's algorithm?
Option A:	24
Option B:	23
Option C:	15
Option D:	19
6.	Bellman Ford algorithm is used to find out single source shortest path for negative edge weights. Bellman Ford algorithm uses which of the following strategy?
Option A:	Greedy method
Option B:	Dynamic Programming
Option C:	Backtracking
Option D:	Divide and Conquer
7.	The optimal solution for 4-queen problem is
Option A:	(2,3,1,4)
Option B:	(1,3,2,4)
Option C:	(3,1,2,4)
Option D:	(2,4,1,3)

8.	Consider the following code snippet:
	Bounding function(k,i) {
	for(j=1 to k-1)
	{ if ((x[j]==i) or (Abs(x[j]-i) ==abs(j-k))) return false;
	} return true }
	The above code represents the bounding function for which of the following algorithm?

Option A:	Subset sum problem using backtracking
Option B:	n-queens using backtracking
Option C:	Graph coloring using backtracking
Option D:	Subset sum using branch and bound
9.	What do you mean by chromatic number?
Option A:	The minimum number of colors needed to color all the vertices optimally in a Graph

10

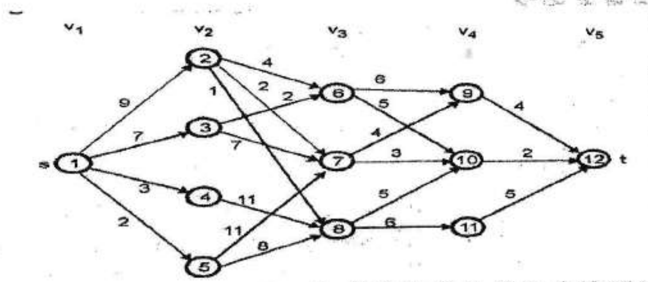
	Coloring problem
Option B:	The maximum number of colors needed to color all the vertices optimally in a Graph Coloring problem
Option C:	The number of colors using which the edges of graph have been colored in a Graph Coloring Problem
Option D:	The individual colors with which we color the vertices of a Graph in a Graph Coloring Problem
10.	Which string matching algorithm uses a Prefix Table?
Option A:	Naïve String Matching Algorithm
Option B:	Boyer Moore String Matching Algorithm
Option C:	Knuth Morris Pratt Algorithm
Option D:	Rabin Karp Algorithm

Q2. (20 Marks)	**Solve any Four out of Six** **05 marks each**
A	Write and Explain binary search algorithm.
B	Write a short note on job sequencing with deadline
C	Determine the LCS of the following sequences: X: {A, B, C, B, D, A, B} Y: {B, D, C, A, B, A}
D	Solve the sum of subsets problem for the following: n=4, m=15, w={3,5,6,7}
E	Give the algorithm for the N-Queen's problem and give any two solutions to the 8-Queen's problem
F	Explain and apply Naïve string matching on following strings String1: COMPANION String2: PANI

Q3. (20 Marks)	**Solve any Two Questions out of Three** **10 marks each**
A	Write algorithm for greedy knapsack and Obtain the solution to following knapsack problem where n=7,m=15 (p1,p2.....p7) = (10,5,15,7,6,18,3), (w1,w2,......,w7) = (2,3, 5,7,1,4,1).
B	Explain Dijkstra's Single source shortest path algorithm. Explain how it is different from Bellman Ford algorithm. Explain 15-puzzle problem using LC search technique.
C	Rewrite and Compare Rabin Karp and Knuth Morris Pratt Algorithms Give the pseudo code for the KMP String Matching Algorithm.

Q4. (20 Marks)	**Solve any Two Questions out of Three** **10 marks each**
A	Write algorithm for quick sort and sort the following elements [40,11,4,72,17,2,49]
B	Write multistage graph algorithm and solve following example.

149

	
C	Write algorithm for 0/1 knapsack problem using dynamic programming .Also solve the following example. N=4, M=21 (p1,p2,p3,p4)=(2,5,8,1),(w1,w2,w3,w4)=(10,15,6,9)

12

Paper / Subject Code: 40522 / Analysis of Algorithm

SE (comp) / Sem IV / R-19 C-Scheme / SH-23 12/12/2023

(3 Hours) Total Marks: 80

QP Code: 39848

N.B: (1) Question No. 1 is compulsory.
(2) Attempt any three from the remaining questions.
(3) Figures to the right indicate full marks.

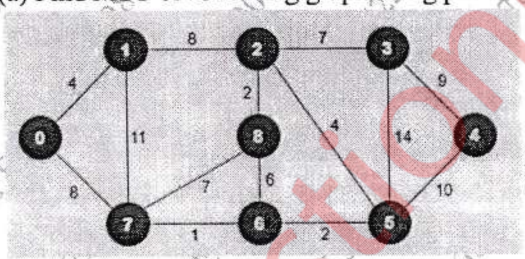
1. Attempt any four (20)
   (a) Explain recurrences and various methods to solve recurrences.
   (b) Explain in brief the concept of Multistage graphs?
   (c) Explain Asymptotic Notations.
   (d) Define P class, NP Class, NP-hard, NP-complete.
   (e) What is greedy algorithm?

2. (a) What is Knuth Morris Pratt Method of Pattern Matching? Give Examples. (10)
   (b) Sort the following numbers using Merge Sort also, derive the time complexity of Merge Sort 7, 2, 9, 4, 3, 8, 6, 1. (10)

3. (a) Explain and differentiate between greedy knapsack and 0/1 knapsack. (10)
   (b) Explain Backtracking with n-queen problem. (10)

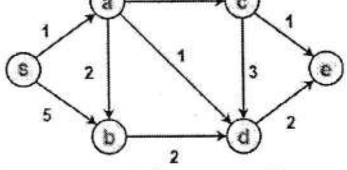
4. (a) Find the LCS for following strings (10)
   String 1- AGGTAB
   String 2- GXTXAYB
   (b) Explain quick sort with algorithm and example. (10)

5. (a) Find MST of following graph using prims and Kruskal's Algorithm. (10)



   (b) Write and explain sum of subset algorithm for n = 5, W = {2, 7, 8, 9, 15} M = 17. (10)

6. Write notes on any two: (20)
   (a) Write an algorithm to find the Minimum and Maximum values using divide and conquer strategy and also derive its complexity.
   (b) Explain Naïve string-matching algorithm with example.
   (c) Find the shortest path from source vertex S using Dijkstra's algorithm.



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(E)

Paper / Subject Code: 40522 / Analysis of Algorithm

COMP/SEM-IV/ R-19 (C-scheme) / QP Code-28902 / 11-05-23

(Time: 3 Hours) Total Marks: 80

N.B: (1) Question No. 1 is compulsory.

(2) Attempt any three from the remaining questions.

(3) Figures to the right indicate full marks.

1. Attempt any four

(A) Describe the relationship along P, NP, NP-hard, NP-complete? 5

(B) What is the difference between divide and conquer approach and dynamic programming? 5

(C) Explain Multistage graph with example. 5

(D) Write an abstract algorithm for greedy design method. 5

(E) What is Asymptotic analysis and define big Oh, big Omega and Theta notation? 5

2. (A) Sort the following numbers using Quick Sort. Also, derive the time complexity of Quick Sort. 50, 31, 71, 38, 77, 81, 12, 33. 10

(B) What is Knuth Morris Pratt Method of Pattern Matching? Give Examples. 10

3. (A) Solve the following instance of Job sequencing with deadlines problem n=7, profits (p1, p2, p3, p4, p5, p6, p7) = (3, 5, 20, 18, 1, 6, 30) and deadlines (d1, d2, d3, d4, d5, d6, d7) = (1, 3, 4, 3, 2, 1, 2). Schedule the jobs in such way so as to get maximum profit. 10

(B) Write and explain sum of subset algorithm for n = 5, W = {2, 7, 8, 9, 15} M = 17. 10

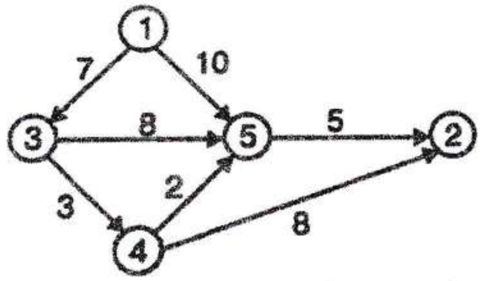
4. (A) Find Longest Common Subsequence for following strings 10

X = acbaed

Y = abcabe

(B) Write an algorithm to find the minimum and maximum value using divide and conquer and also derive its complexity. 10

5. (A) Find a minimum cost path from 3 to 2 in the given graph using dynamic programming. 10



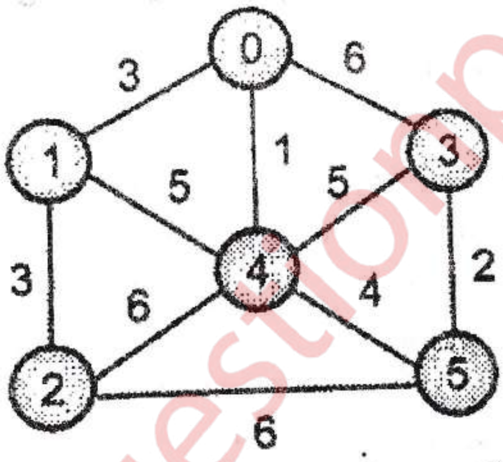
(B) Write an algorithm to solve N Queens problem. Show its working for N = 4. 10

6. Attempt any two 20

(A) Explain naïve string matching algorithm with example.

(B) Explain 0/1 knapsack problem using dynamic programming.

(C) To Find MST of following graph using prim's and kruskal's Algorithm.



University of Mumbai

Examinations Summer 2022

Time: 2 hour 30 minutes | Max. Marks: 80

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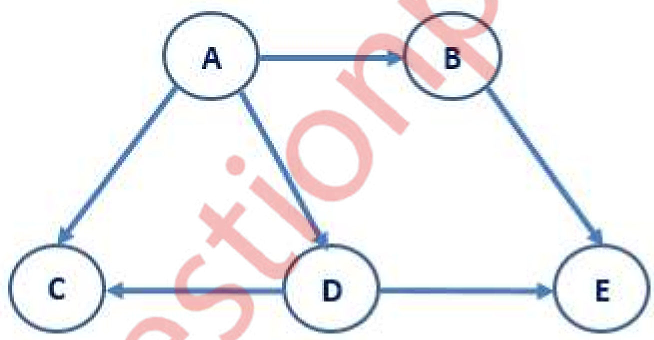
Q1.	**Choose the correct option for following questions. All the Questions are compulsory and carry equal marks**
1.	Consider the following definition in c programming language. Which of the following c code is used to create a new node of circular linked list?

```
struct node
{
        int data;
        struct node *next;
}
typedef struct node NODE;
NODE *ptr;
```

Option A:	ptr = (NODE*)malloc(NODE);
Option B:	ptr = (NODE*)malloc(sizeof(NODE*));
Option C:	ptr = (NODE)malloc(sizeof(NODE));
Option D:	ptr = (NODE*)malloc(sizeof(NODE));
2.	Binary search can be performed, if data items are stored in an
Option A:	Unordered array
Option B:	Ordered array
Option C:	Unordered linked list
Option D:	Ordered linked list
3.	The equivalent postfix expression corresponding to the infix expression (A+B)*(D/C) is
Option A:	ABDC/*+
Option B:	AB+D*C/
Option C:	AB+DC/*
Option D:	ABD*+C/
4.	In the Breadth-First Search traversal of a graph, how many times does a node get visited?
Option A:	Once
Option B:	Twice
Option C:	Equivalent to number of indegree of the node
Option D:	Equivalent to number of outdegree of the node
5.	Linked lists are preferred to other data structures when
Option A:	The elements are in ascending or descending order.
Option B:	No deletion of elements needs to be performed.
Option C:	The number of elements in the list is known beforehand.
Option D:	Insertions and deletions are frequent in a list of unknown sizes.
6.	The number of null links in a binary tree with n nodes is
Option A:	n-1
Option B:	2n – 1
Option C:	2n
Option D:	n + 1
7.	In an AVL tree, difference of height in left sub-tree and right-tree for every node is __________.
Option A:	Zero

Option B:	One
Option C:	Atmost one
Option D:	Atleast one
8.	Suppose a queue is implemented by a circular array QUEUE[0…9]. The number of elements in the queue, if FRONT = 8 and REAR = 3, will be
Option A:	3
Option B:	4
Option C:	5
Option D:	6
9.	_________________ is used in implementation of recursion.
Option A:	Array
Option B:	Stack
Option C:	Queue
Option D:	Tree
10.	In an almost complete binary tree with 13 nodes, the number of leaves will be
Option A:	5
Option B:	6
Option C:	7
Option D:	8

Q2	**Solve any Four out of Six** **5 marks each**
A	Explain different operations that can be performed on data structures.
B	Write a function to delete the last node of the circular linked list.
C	Show the steps for finding the topological sorting of the below graph. 
D	Write an algorithm to evaluate a postfix expression.
E	Write short note on Priority Queue.
F	Construct Binary Search Tree for the following list of elements 45 28 34 63 87 76 31 11 50 17

Q3	**Solve any Two Questions out of Three** **10 marks each**
A	Show the result of inserting 16, 18, 5, 19, 11, 10, 13, 21, 8, 14 one at a time into an initially empty AVL tree.
B	A hash table of size 10 uses linear probing to resolve collisions. The key values are integers and the hash function used is key%10. Draw the table that results after inserting in the given order the following values: 28, 55, 71, 38, 67, 11, 10, 90, 44, 9
C	Write a program to implement Circular queue using an array.

Q4	**Solve any Two Questions out of Three 10 marks each**
A	Write a program to convert the given decimal number to a binary number using stack data structure.
B	Write a program to perform the following operations on a singly linked list i. Insert a new node at the end of the list ii. Delete a node from the beginning of the list iii. Search for a given node iv. Display the list
C	Construct an expression tree for the expression (a + b / c) + ((d * e + f) / g). Give the outputs when you apply preorder and postorder traversals.

Paper / Subject Code: 50923 / Data Structure

SE COMP / SEM-III / R-19 / SH-2022 / 25-11-2022

QP Code- 10012055

(3 Hours) **Total Marks: 80**

N.B: (1) Question No. 1 is compulsory.
(2) Attempt any three questions out of the remaining five questions.
(3) Figures to the right indicate full marks.
(4) Make suitable assumptions wherever necessary.

Q.1 (a) Compare linear and non-linear data structures. [05]

(b) Explain the advantage of circular queue over linear queue. Write a function in C language to insert an element in circular queue. [05]

(c) Define binary search tree. Discuss the case of deletion of a node in binary search tree if node has both the children. [05]

(d) Write a C function to search a node in doubly linked-list. [05]

Q.2 (a) Construct AVL tree for the following sequence: [10]
67,34,90,22,45,11,2,78,37,122

(b) Write algorithm for postfix evaluation. Demonstrate the same step by step for the expression: 9 6 7 * 2 / - [10]

Q.3 (a) Write a program to perform following operations on a circular linked list: [10]
i) insert a node from the end of the list, ii) delete first node,
iii) count the number of nodes with even values, iv) display the list.

(b) Write a C program to simulate linear queue as linked list. [10]

Q.4 (a) Construct Huffman tree and find the Huffman codes for each symbol given below with frequency of occurrence: [10]

Symbol	p	g	e	r	i
Frequency	20	17	33	25	40

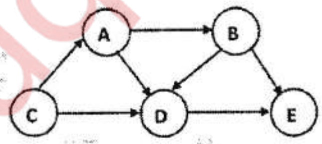
(b) Explain the various ways to represent graph in the memory with example. [05]

(c) Construct binary search tree from given traversal sequences: [05]

In-order traversal	D	E	B	A	C	F	G	I	H	J
Pre-order traversal	F	E	D	C	B	A	G	H	I	J

Q.5 (a) Apply linear probing to hash the following values in a hash table of size 11 and find the number of collisions: 67,44,90,12,83,52,23,87,79. [10]

(b) Define topological sorting. Perform topological sorting for the following graph: [10]



Q.6 (a) Construct a B tree of order 3 by inserting the following given elements as: [10]
77,97,75,64,53,14,26,49,82,59.
Show the B tree at each step of insertion.

(b) Write a function in C for DFS traversal of graph. Explain DFS graph traversal with suitable example. [10]

Paper / Subject Code: 50923 / Data Structure

SE(COMP) / SEM-III / R-19 / C Scheme / SH-23 / 28-11-23

Duration:3 Hours **Total Marks: 80**

QP CODE: 39186

N.B: **(1) Question No. 1 is compulsory.**
(2) Attempt any three questions out of the remaining five questions.
(3) Figures to the right indicate full marks.
(4) Make suitable assumptions wherever necessary with proper justifications.

Q.1. A) Define ADT with an example. [05]

B) Evaluate the postfix expression "94*28+-" using stack ADT. Show the process stepwise. [05]

C) Justify the statement with suitable example: "Circular queue overcomes the disadvantage of linear queue". [05]

D) Differentiate between linear search and binary search. [05]

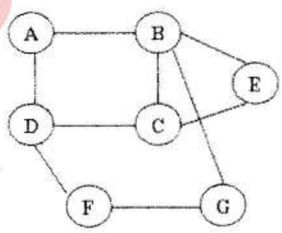
Q.2. A) Construct Huffman tree and determine the code for each symbol in the string "BCAADDDCCACACAC". [10]

B) Discuss the cases of deleting a node from Binary Search Tree with suitable example. [10]

Q.3. A) Write a program in C to implement queue ADT using linked list. [10]

B) Construct an AVL tree by inserting the following elements in the given order. Apply necessary rotations wherever required. [10]
54, 12, 24, 68, 85, 99, 42, 27, 87, 80

Q.4. A) Write C function for BFS graph traversal. Show the stepwise BFS traversal with the help of data structures for the following graph: [10]



B) Write functions in C to perform the following operations on the Doubly Linked List: [10]
i) Delete a node after given node.
ii) Find node with smallest data value.
iii) Display the list.
iv) Insert a node at the end of the list.

Q.5. A) Build a Binary Search Tree, given the following sequences: [05]
Inorder: 35, 41, 48, 52, 57, 72, 79, 85, 86, 90
Preorder: 57, 41, 35, 52, 48, 90, 72, 85, 79, 86

B) What is topological sort? Explain Topological Sorting with an example. [05]

211+10

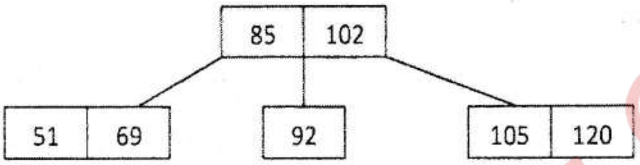
C) What is collision? Using linear probing, insert the following values in the hash table of size 11 & count the no. of collisions: [10]

83, 53, 64, 25, 39, 96, 12,71.

Q.6. A) Write short note on Priority Queue. [05]

B) Write a function in C to count the number of nodes in Singly Linked List. [05]

C) Create a B-tree of order 3 by inserting 87,94,59,98,63,7,27. [10]



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Paper / Subject Code: 50923 / Data Structure

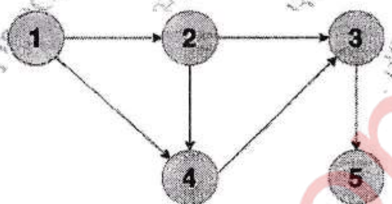
SE(COMP) / SEM III / R-19 / FH-23 / 30-05-23
(ATKT)

QP CODE: 27655

(3 Hours) **Total Marks: 80**

N.B: (1) Question No. 1 is compulsory
(2) Attempt any three questions out of the remaining five questions

Q.1	(a)	Explain various types of data structures with example.	5
	(b)	Define Graph and explainvarious graph representationtechniques.	5
	I	Convert the following expression to postfix. (f-g) * ((a+b) * (c-d))/e	5
	(d)	Differentiate between B tree and B+ tree.	5
Q.2	(a)	Apply linear probing and quadratic probing hash functions to insert values in the Hash table of size 10. Show number of collisions occurs in each technique. 27, 72, 63, 42, 36, 18, 29,101	10
	(b)	Construct B+ tree of order 3 for the following dataset 90, 27, 7, 9, 18, 21, 3, 4, 16, 11, 1, 72	10
Q.3	(a)	Write BFS algorithm. Show BFS traversal for the following graph with all the steps.	10



	(b)	Write a C program to implement linear queue using array.	10
Q.4	(a)	Write a program to perform the following operations on the Singly linked list: i. Insert a node at the end ii. Delete a node from the beginning iii. Search for a given element in the list iv. Display the list	10
	(b)	Write a C program to implement Stack using Linked List	10
Q.5	(a)	Write a program to evaluate postfix expression using stack data structure	10
	(b)	Construct AVL for following elements 50, 25, 10, 5, 7, 3, 30, 20, 8, 15	10
Q.6	(a)	Construct Binary Tree from following traversal. In–order Traversal: D B H E I A F J C G Post order Traversal: D H I E B J F G C A	10
	(b)	Write a C program for polynomial addition using a Linked-list.	10
